FOURTH SEMESTER M.Sc. DEGREE EXAMINATION, JUNE 2018

(CUCSS)

Mathematics

Group IV : MT 4E 14—DIFFERENTIAL GEOMETRY

(2016 Admissions)

me : Three Hours Maximum : 36 Weightage

Part A

Answer **all** *questions.*
Each question has weightage **1.**

1. Describe the level set for the function $f(x_1, x_2) = x_1^2$ at $c = 1$.
2. Sketch the vector field on $\mathbb{R}^2$ given by $X(x_1, x_2) = (x_2, -x_1)$.
3. Show that the n-plane $x_1 + x_2 + \ldots\ldots + x_n + 1 = 1$ is an n-surface.
4. Give an orientation on $\mathbb{R}^2$.
5. Give an example of a surface whose spherical image is S^1.
6. Give an example of a vector field along a parametrized curve.
7. Let α be a parametrized curve on S such that $\alpha' = 0$ along α. Show that α is a geodesic on S.
8. Let X, Y be vector fields parallel along a parametrized curve α. Show that X + Y is also parallel along α.
9. Show that $\alpha \nabla_{cv} f = c \nabla_v f$ where c is a real number and f is a smooth function.
10. Find $L_p(v)$ for $p = (0,0,1)$ and $v = (0, 1, 2)$ on the cylinder $x_2^2 + x_3^2 = 1$ in $\mathbb{R}^3$.
11. Find the length of the parametrized curve $\alpha : [0, \pi] \longrightarrow \mathbb{R}^3$ given by $\alpha(t) = (t, t^2, t^3)$.
12. Find the mean curvature of an n-surface S at p where the principal curvatures are 1, 1/2, 1/4.
13. Give an example of a parametrized 2–surface.
14. Define n-surface in $\mathbb{R}^{n+k}$ for $k > 1$.

(14 × 1 = 14 weightage)

Answer any **seven** *questions.*
Each question has weightage **2.**

15. Sketch the gradient field of the function $f(x_1, x_2) = x_1 - x_2^2$.

16. Show that if $S = f^{-1}(c)$ is an n-surface in $\mathbf{R}^{n+1}$ and $p \in S$ then the tangent space S_p is equal to $(\nabla f(p))^{\perp}$.

17. Find the spherical image of the n-sphere $x_1^2 + x_2^2 + \ldots\ldots + x_{n+1}^2 = 2$.

18. With the usual notations prove that $(X \cdot Y) = \dot{X} \cdot Y + X \cdot \dot{Y}$.

19. Show that $\alpha(t) = (\cos t, \sin t, t)$ is a geodesic on the cylinder $x_1^2 + x_2^2 = 1$ in $\mathbf{R}^3$.

20. Let X' denote the covariant derivative of a vector field X along a parametrized curve α. Show that $(fX)' = f'X + fX'$ for all smooth functions f.

21. Find the curvature k of the plane curve $x_1^2 + x_2^2 = 1, x_1 > 0$.

22. Let $k(p)$ be the curvature of a plane curve at p and v be a non-zero vector tangent to the plane curve at p. Show that $k(p) = L_p(v) \cdot v / \|v\|^2$.

23. Let $N(v)$ be the normal section of an n-surface determined by a unit vector v. Describe an isomorphism from $\mathbb{R}^2$ to $N(v)$.

24. Show that if $f: \mathbb{R}^2 \to \mathbb{R}^4$ is defined by $f(\theta, \phi) = (\cos\theta, \sin\theta, \cos\phi, \sin\phi)$ the image of f is a product of two circles.

$(7 \times 2 = 14$ weightage)

Part C

Answer any **two** *questions.*
Each question has weightage **4.**

25. (a) Define gradient of a smooth function $f: U \longrightarrow \mathbf{R}$ where $U \subseteq \mathbf{R}^{n+1}$.

(b) Let $f: U \longrightarrow \mathbb{R}$ be a smooth function and $p \in U$ with $f(p) = c$. Show that

(i) the gradient of f at p is orthogonal to vectors tangent to $f^{-1}(c)$ at p.

(ii) $(\nabla f(p))^{\perp}$ is equal to the set of all vectors tangent to $f^{-1}(c)$ at p.

26. Let S be a connected n-surface in $\mathbf{R}^{n+1}$. Show that

(a) there exist exactly two normal vector fields on S.

(b) If N_1, N_2 are the two normal vector fields on S then $N_2(p) = -N_1(p)$ for all $p \in S$.

27. Let S be an n-surface in $\mathbf{R}^{n+1}$ and $p, q \in S$.

(a) Define parallel transport from S_p to S_q.

(b) Let $p_\alpha : S_p \longrightarrow S_q$ be a parallel transport. Show that

(i) p_α is a linear map.

(ii) $p_\alpha(v) \cdot p_\alpha(w) = v \cdot w$ for all $v, w \in S_p$.

28. (a) Define second fundamental form on an oriented n-surface.

(b) Let S be a compact connected oriented n-surface in $\mathbf{R}^{n+1}$. Show that the Gauss-Kronecker curvature of S at p is non-zero for all $p \in S$ if and only if the second fundamental form of S at p is definite for all $p \in S$.

(2 × 4 = 8 weightage)